

Einfache Gleichungen und Formelumformungen – Blatt 1

1) Löse die Gleichungen nach der auftretenden Variable:

a) $s + 2 \cdot (1 - 3s) = 2 - s$

b) $1 - (d + 1) = 2 - 3d$

c) $0,1 \cdot (c + 3) = 0,02 + 0,3 \cdot c$

d) $\frac{c}{3} - 1 = 2c - 11$

e) $\frac{k}{2} = \frac{4 - k}{3}$

f) $2 \cdot (1 - 4x) = -(1 + 2x)$

g) $4y - (1 + y) = 0$

h) $2v + 3 = -v + 2$

2) Löse folgende Gleichungen und mache die Probe:

a) $14 - (x - 9) = 10 + 5x - (7x - 5)$

b) $4y - [(3y + 4) + 2] = 3(2y - 8) - (3y - 24)$

c) $(x + 1)(4x - 3) = 2(x + 1)(2x + 3)$

d) $(2x + 1)^2 + (8x - 3)^2 = (7x - 2)(11x - 1) - (3x + 1)^2$

e) $z - 10 = \frac{2z}{3} - 2 - \frac{z}{2}$

f) $24 - \frac{5a}{6} = -\frac{4a}{5} + \frac{5a}{3} - 10$

g) $\frac{b - 3}{6} + \frac{b + 3}{20} = \frac{9}{4}$

h) $\frac{2(a + 1)}{30} + \frac{2a - 10}{5} + 6 = 9 - \frac{9a - 48}{9}$

Lösungen:

1) a) $s = 0$ b) $d = 1$ c) $c = \frac{7}{5}$ d) $c = 6$

e) $k = \frac{8}{5}$ f) $x = \frac{1}{2}$ g) $y = \frac{1}{3}$ h) $v = -\frac{1}{3}$

2) a) $x = -8$ b) $y = -3$ c) $x = -1$ d) $x = 1$

e) $z = \frac{48}{5}$ f) $a = 20$ g) $b = 12$ h) $a = 7$

Einfache Gleichungen und Formelumformungen – Blatt 2

Berechne die gesuchten Variablen:

1 $2a - b = 3(a + b) - c$ $a = ?$

2 $ka + d = b(a + d) + 3$ $a = ?$

3 $a \cdot b = a + b$ $b = ?$

4 $x = \frac{ab}{y - a}$ $b = ?$ $y = ?$ $a = ?$

5 $z + 4 = \frac{a}{b} - z_1$ $a = ?$ $b = ?$

6 $a = b + \frac{b-1}{k-xy}$ $b = ?$ $k = ?$ $x = ?$

7 $\frac{n}{b} = \frac{a}{b} + z$ $a = ?$ $b = ?$

8 $s = s_0 + \frac{v^2}{2a}$ $a = ?$

9 $P = P_1 + \frac{U^2}{R + R_1}$ $R = ?$

10 $R_G = \frac{R_1 \cdot R_2}{R_1 + R_2}$ $R_1 = ?$

11 $f = f_0 \cdot \frac{c}{c - v}$ $v = ?$ $c = ?$

12 $I_1 \cdot R_1 + I_2 \cdot (R_2 + R_1) = U_1$ $R_1 = ?$

Lösungen:

1) $a = c - 4b$

2) $a = \frac{bd + 3 - d}{k - b}$

3) $b = \frac{a}{a - 1}$

4) $b = \frac{x(y - a)}{a}$ / $a = \frac{xy}{b + x}$ / $y = \frac{ab}{x} + a = \frac{ab + ax}{x}$

5) $a = b(z + 4 + z_1)$ / $b = \frac{a}{z + 4 + z_1}$

6) $b = \frac{a(k - xy) + 1}{k - xy + 1}$ / $k = \frac{b - 1 + xy(a - b)}{a - b} = \frac{b - 1}{a - b} + xy$

$x = \frac{(a - b)k - b + 1}{y(a - b)} = \frac{k}{y} - \frac{b - 1}{y(a - b)}$

7) $a = n - bz$ / $b = \frac{n - a}{z}$

8) $a = \frac{v^2}{2(s - s_0)}$

9) $R = \frac{U^2 - R_1(P - P_1)}{P - P_1} = \frac{U^2}{P - P_1} - R_1$

10) $R_1 = \frac{R_2 \cdot R_G}{R_2 - R_G}$

11) $v = \frac{c(f - f_0)}{f}$ / $c = \frac{vf}{f - f_0}$

12) $R_1 = \frac{U_1 - I_2 R_2}{I_1 + I_2}$

Einfache Gleichungen und Formelumformungen – Blatt 3

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| <p>1) $G = \frac{E}{2 \cdot (1+f)}$ ges.: f, E</p> | <p>2) $p_0 + \frac{1}{2} \rho v_1^2 = p$ ges.: ρ, p_0</p> |
| <p>3) $v = \frac{p_1 - p_2}{4\eta l} (r^2 - x^2)$ ges.: p_2, l</p> | <p>4) $y = A f \cdot \left(t - \frac{r}{c} \right)$ ges.: r, c</p> |
| <p>5) $v_1 = -c + \frac{2\pi r c}{\lambda}$ ges.: c, λ</p> | <p>6) $\left(p + \frac{a}{V^2} \right) \cdot (V - b) = RT$ ges.: b, a</p> |
| <p>7) $c_p = c_v + \frac{R}{M}$ ges.: M, R</p> | <p>8) $\Delta T = \frac{V'' - V'}{Q} \cdot \Delta p$ ges.: V', Q</p> |
| <p>9) $U_2 = \frac{UR_2}{R_1 + R_2}$ ges.: R_2, R_1</p> | <p>10) $I \cdot R_1 = R_3 \cdot \left(i - i' \frac{R_2}{R_3} \right)$ ges.: i', R_3</p> |
| <p>11) $K = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2}$ ges.: Q_1, ϵ_0</p> | <p>12) $\eta = \frac{1}{1 + 4\pi\kappa}$ ges.: κ</p> |
| <p>13) $l = \frac{4\pi\kappa H}{1 + \beta\kappa}$ ges.: κ, β</p> | <p>14) $c_1 = c + v \left(1 - \frac{1}{n^2} \right)$ ges.: v, c</p> |
| <p>15) $E = h(f_1 - f_2) + mc^2$ ges.: f_2, h</p> | <p>16) $\frac{4\pi}{3} r^3 (\rho - \rho') g = 6\pi\eta r v$ ges.: ρ, ρ'</p> |
| <p>17) $c = \frac{(c_1 m_1 + c_2 m_2)(t - t_1)}{m(t_2 - t)}$ ges.: t</p> | <p>18) $\left(1 + \frac{\lambda_1}{K_1} \right) B_1 = \left(1 + \frac{\lambda_2}{K_2} \right) B_2$ ges.: K_2</p> |

Lösungen:

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|---|--|
| <p>1) $f = \frac{E - 2G}{2G}$; $E = 2G(f + 1)$</p> | <p>2) $\rho = \frac{2(p - p_0)}{v_1^2}$; $p_0 = \frac{2p - \rho v_1^2}{2}$</p> |
| <p>3) $p_2 = \frac{p_1(r^2 - x^2) - 4\eta l v}{r^2 - x^2}$; $l = \frac{(r^2 - x^2)(p_1 - p_2)}{4\eta v}$</p> | <p>4) $r = \frac{c(AfT - y)}{Af}$; $c = \frac{Afr}{Aft - y}$</p> |
| <p>5) $c = \frac{\lambda v_1}{2\pi r - \lambda}$; $\lambda = \frac{2\pi cr}{c + v_1}$</p> | <p>6) $b = \frac{V(a + pV^2 - RTV)}{a + pV^2}$; $a = \frac{V^2(bp + RT - pV)}{V - b}$</p> |
| <p>7) $M = \frac{R}{c_p - c_v}$; $R = M(c_p - c_v)$</p> | <p>8) $V' = \frac{V''\Delta p - Q\Delta T}{\Delta p}$; $Q = \frac{V'' - V'}{\Delta T} \Delta p$</p> |
| <p>9) $R_2 = \frac{R_1 U_2}{U - U_2}$; $R_1 = \frac{R_2(U - U_2)}{U_2}$</p> | <p>10) $i' = \frac{iR_3 - IR_1}{R_2}$; $R_3 = \frac{i'R_2 + IR_1}{i}$</p> |
| <p>11) $Q_1 = \frac{4\pi\epsilon_0 r^2 K}{Q_2}$; $\epsilon_0 = \frac{Q_1 Q_2}{4\pi r^2 K}$</p> | <p>12) $\kappa = \frac{1 - \eta}{4\pi\eta}$</p> |
| <p>13) $\kappa = \frac{l}{4\pi H - l\beta}$; $\beta = \frac{4\pi H\kappa - l}{l\kappa}$</p> | <p>14) $v = \frac{n^2(c_1 - c)}{n^2 - 1}$; $c = \frac{n^2 c_1 + v(1 - n^2)}{n^2}$</p> |
| <p>15) $f_2 = \frac{hf_1 + mc^2 - E}{h}$; $h = \frac{mc^2 - E}{f_2 - f_1}$</p> | <p>16) $\rho = \frac{2gr^2\rho' + 9v\eta}{2gr^2}$; $\rho' = \frac{2g\rho r^2 - 9v\eta}{2gr^2}$</p> |
| <p>17) $t = \frac{(m_1 c_1 + m_2 c_2)t_1 + m c t_2}{m_1 c_1 + m_2 c_2 + mc}$</p> | <p>18) $K_2 = \frac{B_2 K_1 \lambda_2}{B_1(K_1 + \lambda_1) - B_2 K_1}$</p> |